🏆 Infosys SE Elite Crack Vault — 09 Mensuration 2D 3D and Coordinate Geometry Elite

🎯 Infosys Systems Engineer (SE) | 📊 10 High-Yield Questions | 🗓️ 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 01 • Cube Root & Mensuration Shortcuts

🔥 MUST-SOLVE

ID: SEC02_BOOK_P05_Q30 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

Find the cube root of 328,509:
$$\sqrt[3]{328509}$$

- a. 69
- b. 50
- c. 97
- d. 78

✓ **Verified Answer:** A. 69

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Units Digit Determination:**
   The number ends in 9. The only single-digit number whose cube ends in 9 is 9 ($9^3 = 729$).
   $$\text{Units digit of } \sqrt[3]{328509} = 9$$
2. **Tens Digit Determination:**
   Strike out the last three digits (509), leaving the prefix number 328.
   Determine the largest integer n such that $n^3 \le 328$:
   $$6^3 = 216 \le 328 < 7^3 = 343$$
   Therefore, the tens digit is **6**.
3. **Verification:**
   $$69^3 = 69 \times 69 \times 69 = 4,761 \times 69 = 328,509$$
   Therefore, the cube root is **69**. Verified Answer is **A**.

Problem 02 • Cube Root & Mensuration Shortcuts

MUST-SOLVE

ID: SEC02_BOOK_P05_Q31 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

Find the cube root of 50,653:
$$\sqrt[3]{50653}$$

- a. 22
- b. 37
- c. 97
- d. 69

✓ **Verified Answer:** B. 37

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Units Digit Determination:**
   The number ends in 3. The only single-digit number whose cube ends in 3 is 7 ($7^3 = 343$).
   $$\text{Units digit of } \sqrt[3]{50653} = 7$$
2. **Tens Digit Determination:**
   Strike out the last three digits (653), leaving the prefix number 50.
   Determine the largest integer n such that $n^3 \le 50$:
   $$3^3 = 27 \le 50 < 4^3 = 64$$
   Therefore, the tens digit is **3**.
3. **Verification:**
   $$37^3 = 37 \times 37 \times 37 = 1,369 \times 37 = 50,653$$
   Therefore, the cube root is **37**. Verified Answer is **B**.

Problem 03 • Angular Geometry & Directional Rotation

MUST-SOLVE

ID: SEC02_BOOK_P26_Q199 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

Shiva is facing North. He turns 180° and 45° in the clockwise direction, then 270° in the anti-clockwise direction, and finally turns 180°. What is his facing direction now?

- a. SE (South-East)
- b. SW (South-West)
- c. South
- d. NE (North-East)

✓ **Verified Answer:** A. SE (South-East)

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Angular Reference Frame:**
   Let North be 0° (or 360°). Clockwise (CW) rotations are positive ($+$) and Anti-Clockwise (ACW) rotations are negative ($-$).
2. **Net Angular Calculation:**
   $$\text{Initial Position} = 0^\circ \text{ (North)}$$
   $$\Delta \theta_1 = +180^\circ \text{ (CW)}$$
   $$\Delta \theta_2 = +45^\circ \text{ (CW)}$$
   $$\Delta \theta_3 = -270^\circ \text{ (ACW)}$$
   $$\Delta \theta_4 = +180^\circ \text{ (CW or general reverse)}$$
3. **Sum of Angles:**
   $$\theta_{\text{final}} = 0^\circ + 180^\circ + 45^\circ - 270^\circ + 180^\circ = 135^\circ$$
4. **Compass Mapping:**
5. 0° = North
6. 90° = East
7. 135° = South-East (SE)
8. 180° = South
   Therefore, Shiva is facing **South-East (SE)**. Verified Answer is **A**.

Problem 04 • 2D Perimeter & Wire Re-bending Invariant

MUST-SOLVE

ID: SEC02_MENSUR_WIRE_Q01 | Source: CampusMonk Quantitative Aptitude | Level: Infosys SE Standard

A copper wire is bent in the form of a circle of radius 28 cm. If the same wire is straightened and re-bent into the shape of a square, what is the length of each side of the square? (Take $\pi = 22/7$)

- a. 40 cm
- b. 44 cm
- c. 48 cm
- d. 52 cm

✓ **Verified Answer:** B. 44 cm

📐 **STEP-BY-STEP DERIVATION & VERIFICATION PROOF**

1. **Length Conservation Invariant:**
   When a wire of fixed length is re-bent from one shape to another, the perimeter remains strictly invariant:
   $$\text{Perimeter of Square} = \text{Circumference of Circle}$$
2. **Circumference of the Circle:**
   $$C = 2\pi r = 2 \times \frac{22}{7} \times 28 = 2 \times 22 \times 4 = 176\text{ cm}$$
3. **Side of the Square (a):**
   $$4a = 176 \implies a = \frac{176}{4} = 44\text{ cm}$$
   Therefore, the length of each side of the square is **44 cm**. Verified Answer is **B**.

Problem 05 • 3D Cuboid Surface Area & Ratio Scaling

MUST-SOLVE

ID: SEC02_TEST_MENSUR_Q01 | Source: Mensuration Pre-Test_QUESTIONS.json | Level: Infosys SE Standard

The ratio of the length, width, and height of a closed rectangular cuboid is $7 : 5 : 4$. If its entire surface area (Total Surface Area) is $2,656\text{ cm}^2$, find its width.

- a. 16 cm
- b. 20 cm
- c. 24 cm
- d. 28 cm

✓ **Verified Answer:** B. 20 cm

📐 STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Dimension Modeling:**
   Let the dimensions of the cuboid be:
   $$\text{Length } l = 7x, \quad \text{Width } w = 5x, \quad \text{Height } h = 4x$$
2. **Total Surface Area (TSA) Formula:**
   $$\text{TSA} = 2(lw + wh + hl)$$
   $$\text{TSA} = 2[(7x \times 5x) + (5x \times 4x) + (4x \times 7x)]$$
   $$\text{TSA} = 2[35x^2 + 20x^2 + 28x^2] = 2[83x^2] = 166x^2$$
3. **Equating to Given Area:**
   $$166x^2 = 2656$$
   $$x^2 = \frac{2656}{166} = 16 \implies x = \sqrt{16} = 4$$
4. **Calculating Width (w):**
   $$w = 5x = 5 \times 4 = 20\text{ cm}$$
   (Also: length $= 7 \times 4 = 28\text{ cm}$, height $= 4 \times 4 = 16\text{ cm}$).
   Therefore, the width is **20 cm**. Verified Answer is **B**.

Problem 06 • Cylinder Total Surface Area

CORE SE

ID: SEC02_TEST_MENSUR_Q02 | Source: Mensuration Pre-Test_QUESTIONS.json | Level: Infosys SE Standard

Find the total surface area (in cm^2) of a right circular cylinder having a diameter of 7 cm and a height of 6 cm. (Take $\pi = 22/7$)

- a. 185 cm^2
- b. 198 cm^2
- c. 209 cm^2
- d. 220 cm^2

✓ **Verified Answer:** C. 209 cm^2

STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Given Parameters:**
2. Diameter $d = 7\text{ cm} \implies \text{Radius } r = \frac{7}{2} = 3.5\text{ cm}$
3. Height $h = 6\text{ cm}$
4. **Formula for Total Surface Area of Cylinder:**
   $$\text{TSA} = 2\pi r(r + h)$$
5. **Arithmetic Substitution:**
   $$\text{TSA} = 2 \times \frac{22}{7} \times \frac{7}{2} \times \left(\frac{7}{2} + 6 \right)$$
   $$\text{TSA} = 22 \times \left(3.5 + 6 \right) = 22 \times 9.5 = 209\text{ cm}^2$$
   Therefore, the total surface area is **209 cm^2**. Verified Answer is **C**.

Problem 07 • 3D Sphere Melting & Recasting Conservation

CORE SE

ID: SEC02_MENSUR_RECAST_Q01 | Source: CampusMonk Quantitative Aptitude | Level: Infosys SE Standard

A solid metallic sphere of radius 6 cm is melted and recast into small spherical lead balls, each of radius 2 cm. How many such small balls can be made from the parent sphere?

- a. 18
- b. 24
- c. 27
- d. 36

✓ **Verified Answer:** C. 27

STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Volume Conservation Invariant:**
   When a metal body is melted and recast, total volume is conserved:
   $$V_{\text{parent}} = n \times V_{\text{small}}$$
2. **Volume of a Sphere Formula:**
   $$V = \frac{4}{3}\pi R^3$$
3. **Ratio of Volumes:**
   $$n = \frac{\frac{4}{3}\pi R^3}{\frac{4}{3}\pi r^3} = \left(\frac{R}{r} \right)^3$$
4. **Substitution:**
   $$n = \left(\frac{6}{2} \right)^3 = 3^3 = 27$$
   Therefore, exactly **27** small spherical balls can be manufactured. Verified Answer is **C**.

Problem 08 • Cone and Cylinder Equal Base Volume Comparison

CORE SE

ID: SEC02_MENSUR_CONE_CYL_Q01 | Source: CampusMonk Quantitative Aptitude | Level: Infosys SE Standard

A right circular cone and a cylinder have equal circular bases and identical heights. If the volume of the cylinder is 180 cm^3, what is the volume of the cone?

- a. 45 cm^3
- b. 60 cm^3
- c. 90 cm^3
- d. 120 cm^3

✓ **Verified Answer:** B. 60 cm^3

STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Geometric Relationship:**
   For identical base radius r and identical height h:
   $$V_{\text{cylinder}} = \pi r^2 h$$
   $$V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3} V_{\text{cylinder}}$$
2. **Evaluation:**
   $$V_{\text{cone}} = \frac{1}{3} \times 180 = 60\text{ cm}^3$$
   Therefore, the volume of the cone is **60 cm^3**. Verified Answer is **B**.

Problem 09 • Circular Track Concentric Pathway Area

CORE SE

ID: SEC02_MENSUR_TRACK_Q01 | Source: CampusMonk Quantitative Aptitude | Level: Infosys SE Standard

A circular garden of radius 21 m is surrounded by a running track of uniform width 3.5 m. What is the area of the running track? (Take $\pi = 22/7$)

- a. 462.5 m^2
- b. 485.0 m^2
- c. 500.5 m^2
- d. 525.0 m^2

✓ **Verified Answer:** C. 500.5 m^2

STEP-BY-STEP DERIVATION & VERIFICATION PROOF

1. **Concentric Radii:**
2. Inner radius $r = 21\text{ m}$
3. Outer radius $R = r + \text{width} = 21 + 3.5 = 24.5\text{ m}$
4. **Annulus (Track) Area Formula:**
   $$\text{Area} = \pi (R^2 - r^2) = \pi (R - r)(R + r)$$
5. **Calculation:**
   $$\text{Area} = \frac{22}{7} \times (24.5 - 21) \times (24.5 + 21)$$
   $$\text{Area} = \frac{22}{7} \times 3.5 \times 45.5$$
   Since $\frac{3.5}{7} = 0.5$:
   $$\text{Area} = 22 \times 0.5 \times 45.5 = 11 \times 45.5 = 500.5\text{ m}^2$$
   Therefore, the area of the track is **500.5 m^2**. Verified Answer is **C**.

Problem 10 • Trapezium Field Geometry

CORE SE

ID: SEC02_MENSUR_TRAP_Q01 | Source: CampusMonk Quantitative Aptitude | Level: Infosys SE Standard

The parallel sides of an isosceles trapezium are 20 cm and 10 cm. Each of the non-parallel oblique sides is 13 cm. What is the total area of the trapezium?

- a. 150 cm^2
- b. 180 cm^2
- c. 210 cm^2
- d. 240 cm^2

✓ **Verified Answer:** B. 180 cm^2

📐 **STEP-BY-STEP DERIVATION & VERIFICATION PROOF**

1. **Determining Height of Trapezium (h):**
   In an isosceles trapezium with parallel sides $a = 20\text{ cm}$ and $b = 10\text{ cm}$, dropping perpendiculars from the top corners to the base leaves two symmetric right triangles with base:
   $$x = \frac{a - b}{2} = \frac{20 - 10}{2} = 5\text{ cm}$$
2. **Applying Pythagorean Theorem for Height (h):**
   The hypotenuse is the non-parallel side $= 13\text{ cm}$.
   $$h = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12\text{ cm}$$
3. **Trapezium Area Formula:**
   $$\text{Area} = \frac{1}{2} \times (a + b) \times h$$
   $$\text{Area} = \frac{1}{2} \times (20 + 10) \times 12 = 15 \times 12 = 180\text{ cm}^2$$
   Therefore, the area of the trapezium is **180 cm^2**. Verified Answer is **B**.